MIL-R-22928B

29 March 1974
SUPERSEDING
MIL-R-22928A
17 September 1964

MILITARY SPECIFICATION

RECEIVERS, LIQUID REFRIGERANT 12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid receiver for use with refrigerant 12.

1.2 Classification. Receivers shall be of the following sizes as specified (see 6.2):

Size	Minimum Capacity Cubic Inches
1	200
2	350
3	450
4	550
5	1300

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants
PPP-B-636 - Boxes, Shipping, Fiberboard

FSC 4130

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MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 80" Pallets, Skids, Runners, or Pallet Type Base
MS 17291 - Receivers, Liquid, Refrigerant 12

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society of Mechanical Engineers (ASME)

Boiler and Pressure Vessel Code. Pressure Vessels Section VIII, Division 1

(Application for copies should be addressed to the American Society of Mechanical Engineers United Engineering Center, 345 East 47th Street, New York, NY 10017.)

Underwriters' Laboratories, Inc. (UL)

Standard 207C - Refrigerant Containing Components

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, I 60062; 1285 Walt Whitman Road, Melville, L.I., New York 11746; or 1655 Scott Blvd., Santa Clara, CA 95050.)

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Standard product. The receivers shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

* 3.2 Codes and standards. The sizes 1, 2, 3, and 4 receivers shall comply with the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standard 207C or American Society of Mechanical Engineers (ASME), Boiler and Pressure Vessel Code, Pressure Vessels, section VIII, division 1. The size 5 receivers shall comply with the American Society of Mechanical Engineers (ASME) Code of Unfired Pressure Vessels, section VIII.

* 3.2.1 Compliance. Prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the sizes 1, 2, 3, and 4 receivers he proposes to furnish under this specification meet the applicable requirements of UL 207C or ASME Code of Unfired Pressure Vessels, section VIII, and the size 5 receivers meet the requirements of ASME Code of Unfired Pressure Vessels, section VIII.

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* 3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the sizes 1, 2, 3, and 4 receivers have been tested and conform to UL Standard No. 207C.

* 3.2.1.2 ASME. Acceptable evidence of meeting the requirements of ASME shall be the ASME code symbol or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the receivers have been tested and conform to ASME Code of Unfired Pressure Vessels, section VIII.

* 3.3 Materials and components. Materials and components shall be as specified herein and in MS 17291. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial receiver provided the completed item complies with all provisions of this specification.

* 3.4 Design and construction. The receiver shall be of the horizontal, cylindrical type, designed and constructed in accordance with MS 17291 and suitable for storage of liquid refrigerant 12, conforming to type 12 of BB-F-1421. The size 1, 2, 3 and 4 receivers shall be of welded or brazed-steel construction and the size 5 shall be welded-steel construction. Receivers shall be furnished complete with plugged bosses for attachment of inlet and outlet shutoff valves and a pressure relief device or fusible plug. The ends shall be stamped or spun and welded or brazed for sizes, 1, 2, 3, and 4 and welded for size 5, to the cylindrical barrel. The interior of the receivers shall be blown clean of foreign matter and dehydrated, and shall have oxygen eliminated prior to plugging by creating an internal atmosphere of dry nitrogen or dry air with a minus 40 F. dew point.

3.5 Marking.

3.5.1 Identification. Unless otherwise specified (see 6.2) in addition to any marking specified by UL or ASME, the letters, "US" shall be clearly etched or embossed on the receiver. In addition, the manufacturer's name or easily identifiable trade name, and year of manufacture shall be clearly stamped, embossed, or etched on a securely attached corrosion-resistant metal plate. In lieu of the metal identification plate a pressure-sensitive label or waterproofed decalcomania attached to the shell with an adhesive may be used.

3.5.2 Special markings. The outlet connection shall be marked with the word "outlet" on the boss.

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* 3.6 Workmanship. All components and assemblies of the receivers shall be free from dirt and other harmful extraneous materials, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow-holes and sprues. External surfaces shall be free from sharp edges and corners, except where corners are required.

* 3.6.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn-holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish weld area.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspoected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.2.2 Process examination. Examination shall be made to insure receivers are cleaned and dehydrated, charged and plugged (see 3.4), to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

* 4.2.3 End item inspection. The inspection lot shall be all receivers of one size offered for inspection at one time.

* 4.2.3.1 Visual examination. The receivers shall be examined for the defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	Not manufacturer's standard commercial product	X	
Design, construction and workmanship	Fractured, split, punctured, dented, bowed, malformed	X	
	Component missing, broken, or threads damaged	X	
Welding	Not smooth or free from cracks, burn-holes or undercuts	X	
	Undercut or incomplete fusion	X	
	Not free from rust, scale, paint, grease or foreign matter		X
Plugging	Not properly accomplished or missing	X	
Identification	Missing, incomplete, not legible, not specified type plate, not fabricated of specified material, not etched, stamped, embossed, printed, or lithographed, whichever required		X
Special marking	Outlet not marked as specified	X	
	Marking required by referenced specifications and standards missing	X	

* 4.2.3.2 Dimensional examination. The receivers shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

* 4.2.4 Inspection of preparation for delivery. An inspection will be made to determine that packaging, packing, and marking, as required by section 5 of this specification, are complied with. Defects shall be scored as set forth in table II. The inspection lot shall be the number of containers fully prepared for inspection. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

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* TABLE II. Examination of preparation for delivery

Examination	Defect
Marking, exterior and interior	Omitted; incorrect; illegible, of improper size, location, sequence, or method of application
Material	Component missing, damaged, or otherwise defective
Contents	Number per container is more or less than required

4.2.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with PPP-B-636.

* 4.2.5 Certificate examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each receiver shall be preserved and packaged method III of MIL-P-116 by packaging in a snug-fitting fiberboard box conforming to style FOL end-opening, type CF or SF, class comestic, variety SW, grade 200 of PPP-B-636. Fiberboard pads or inserts shall be provided as required to prevent the receiver connections from puncturing the box during shipment or storage. Closure shall be in accordance with the appendix of the box specification.

* 5.1.2 Level C. Receivers of the same size shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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* 5.2.1 Level A. Twelve size 1, eight size 2, six size 3, six size 4 or three size 5 receivers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FTC, V2s of PPP-B-636, except that the weight limitation shall be waived. Each shipping container shall be closed and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification.

* 5.2.2 Level B. Twelve size 1, eight size 2, six size 3, six size 4 or three size 5 receivers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FTC, type CF, variety DW, or type SF, class domestic, grade 350 of PPP-B-636, except that the weight limitation shall be waived. Each shipping container shall be reinforced as specified in 5.2.1.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s fabricated and closed in accordance with PPP-B-636.

* 5.2.3 Level C. Receivers, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

* 5.4 Palletization. Unless otherwise specified (See 6.2), receivers packed as specified, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

6. NOTES

6.1 Intended use. The receivers covered by this specification are intended for use in refrigeration systems for the storage of liquified Refrigerant 12.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Whether U.S. marking is required (see 3.5.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is not required (see 5.4).

* 6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) were made from previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Navy - YD

Reviwe activities:

Army - MD, ME
DGSC - SS

User activity:

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Preparing activity:

Army - GL

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-R-22928B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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